Northern University of Business and Technology Khulna

Department of Computer Science and Engineering

# Project Title

## MediLink – Smart Doctor Appointment & Healthcare Management System.

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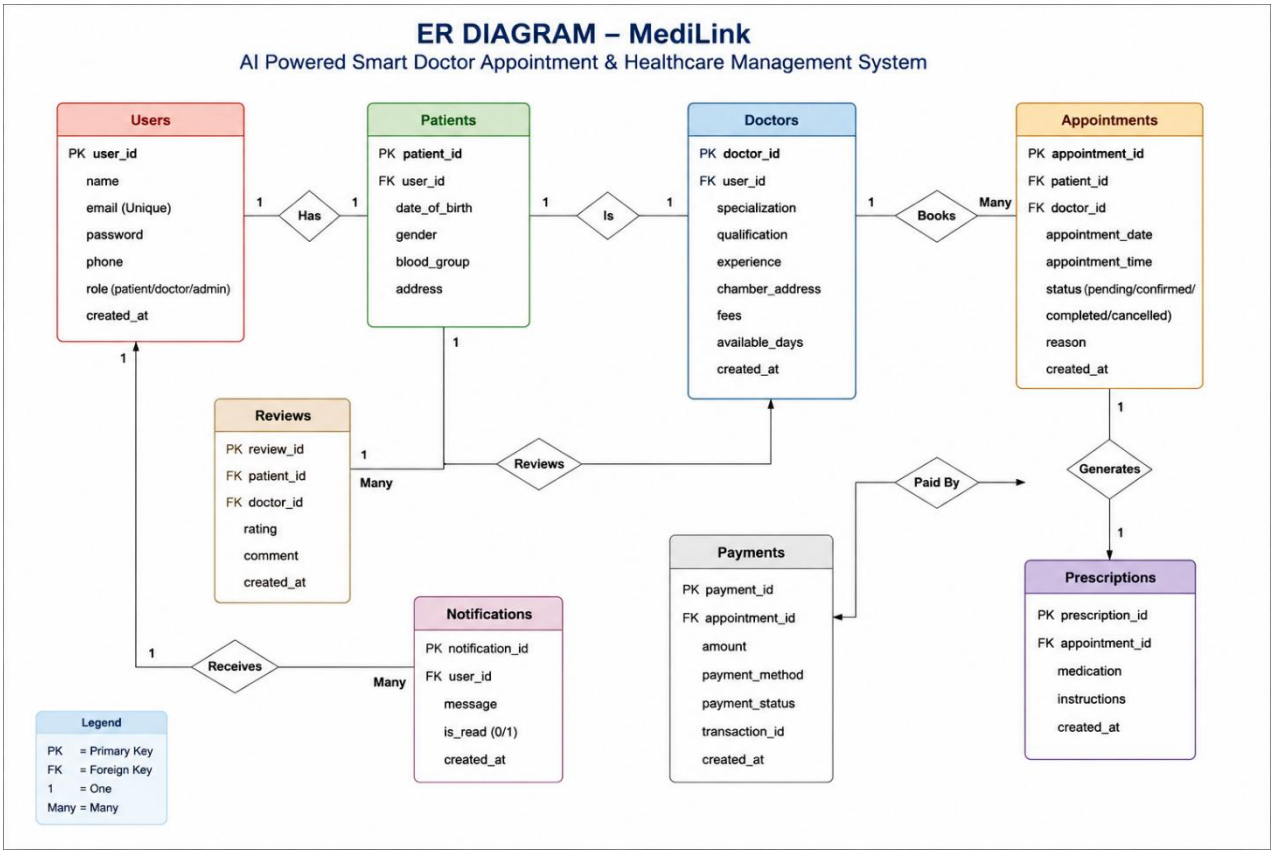
# Database Design and Architecture Description of MediLink

## ER Diagram

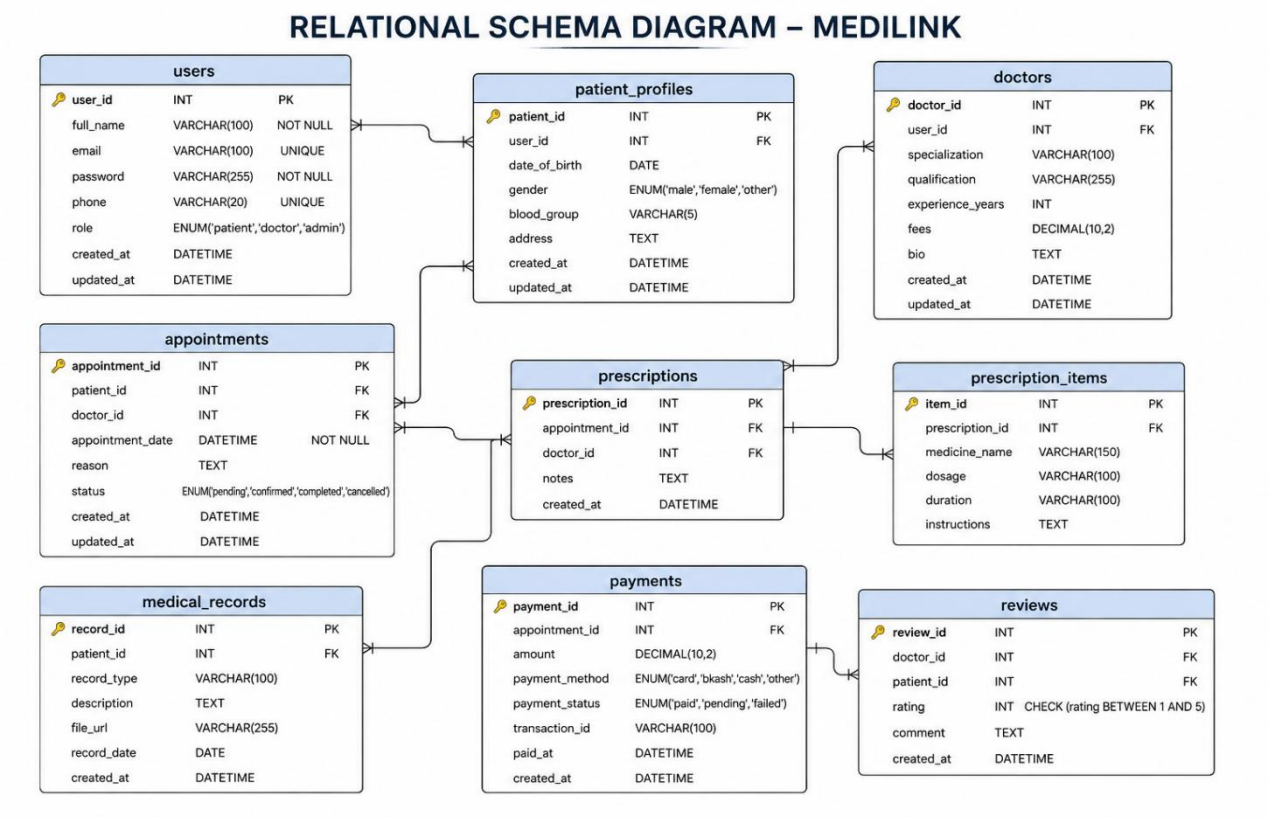



## Description

The Entity Relationship (ER) Diagram of MediLink illustrates the major entities of the AI-Powered Smart Doctor Appointment and Healthcare Management System and the relationships among them.The primary entities in the system are Users, Patients, Doctors, Appointments, Prescriptions, Payments, Reviews, and Notifications.The Users entity stores general information about all system users, including name, email, password, phone number, role, and account creation date. A user can be registered as a patient, doctor, or administrator.The Patients entity contains patient-specific information such as date of birth, gender, blood group, and address. Similarly, the Doctors entity stores professional details including specialization, qualifications, experience, chamber address, consultation fees, and available working days.A patient can book multiple appointments with doctors through the Appointments entity. Each appointment contains appointment date, time, status, and the reason for consultation.After an appointment is completed, the doctor may generate a Prescription containing medications and treatment instructions. Appointment-related payments are recorded in the Payments entity,

which stores payment amount, method, status, and transaction details.Patients can provide feedback and ratings for doctors through the Reviews entity. In addition, the Notifications entity manages system-generated messages and alerts sent to users regarding appointments, prescriptions, and other healthcare activities. Overall, the ER Diagram provides a clear representation of how different entities interact within the MediLink healthcare ecosystem.

## Relational Schema Diagram




## Description

The Relational Schema Diagram represents the logical database structure of the MediLink system. It defines all tables, their attributes, primary keys, foreign keys, and data types.

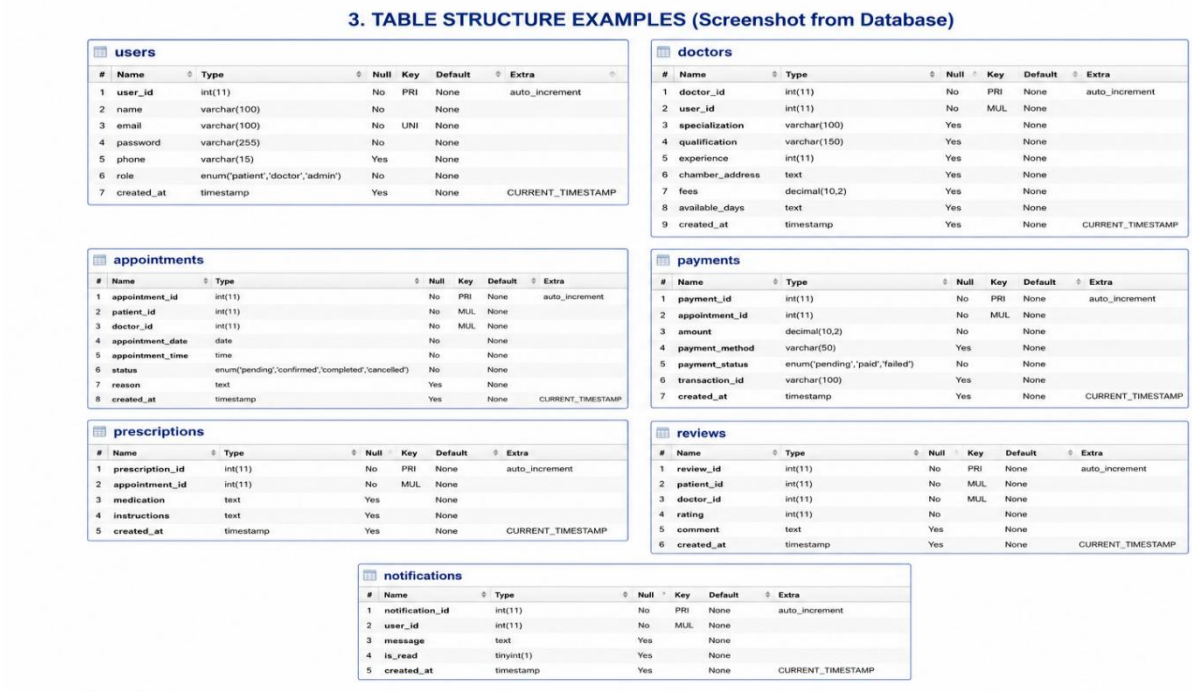
**The database consists of the following major tables:**

- users
- patient_profiles
- doctors
- appointments

- prescriptions
- prescription_items
- medical_records
- payments
- reviews

Each table is designed to maintain data integrity and minimize redundancy through normalization.The users table serves as the central table that manages authentication and user information. The patient_profiles and doctors tables extend user information by storing patient-specific and doctor-specific details.The appointments table connects patients and doctors and stores appointment scheduling information. The prescriptions table records prescriptions generated after consultations, while prescription_items stores detailed medicine information such as dosage, duration, and instructions.The medical_records table maintains patient health records and uploaded documents. The payments table manages transaction details for appointments, and the reviews table stores patient feedback and doctor ratings.The schema establishes relationships through foreign keys, ensuring referential integrity and efficient database management.

## Table Structure Examples



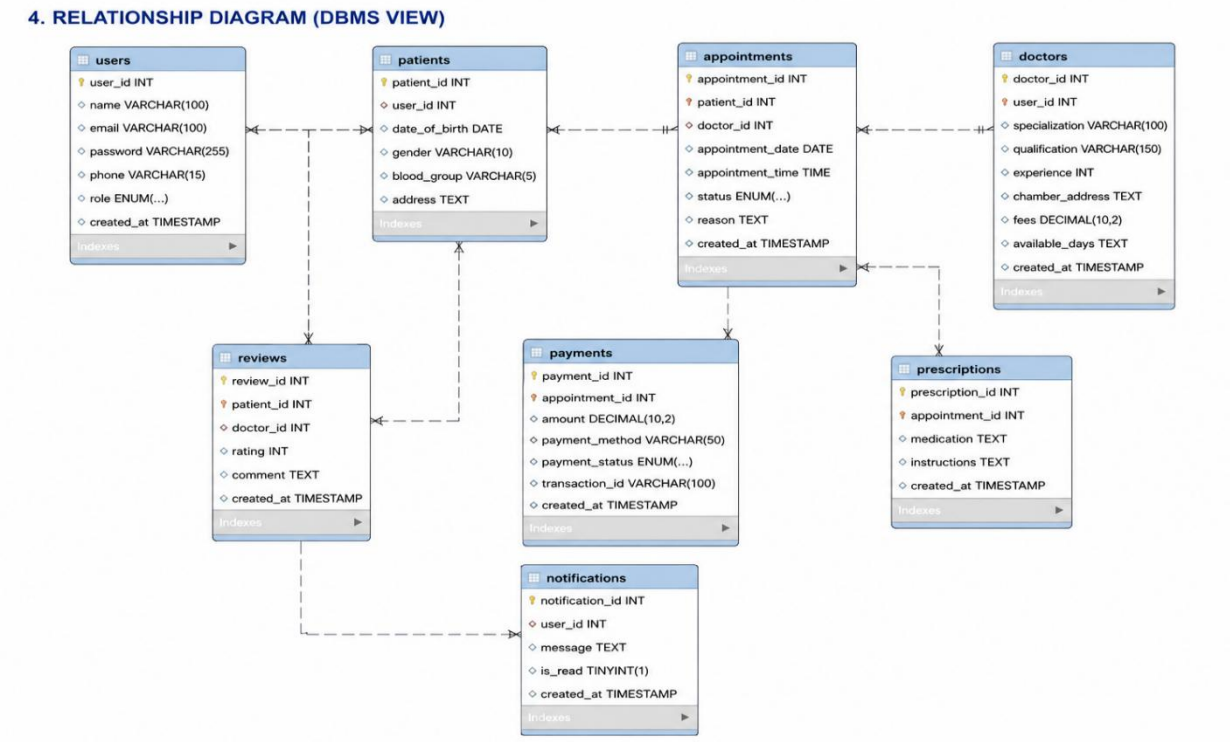
## Description

The Table Structure Examples section presents screenshots of actual database tables created in the MySQL database.The screenshots demonstrate the implementation of the system's database schema and display the following information:

- Column names
- Data types
- Primary keys
- Foreign keys
- Null constraints
- Default values
- Auto-increment properties

Examples include the Users, Doctors, Appointments, Payments, Prescriptions, Reviews, and Notifications tables.These screenshots verify that the database design has been successfully implemented according to the proposed schema and provide practical evidence of the database structure used in the MediLink system.

## Relationship Diagram (DBMS View)



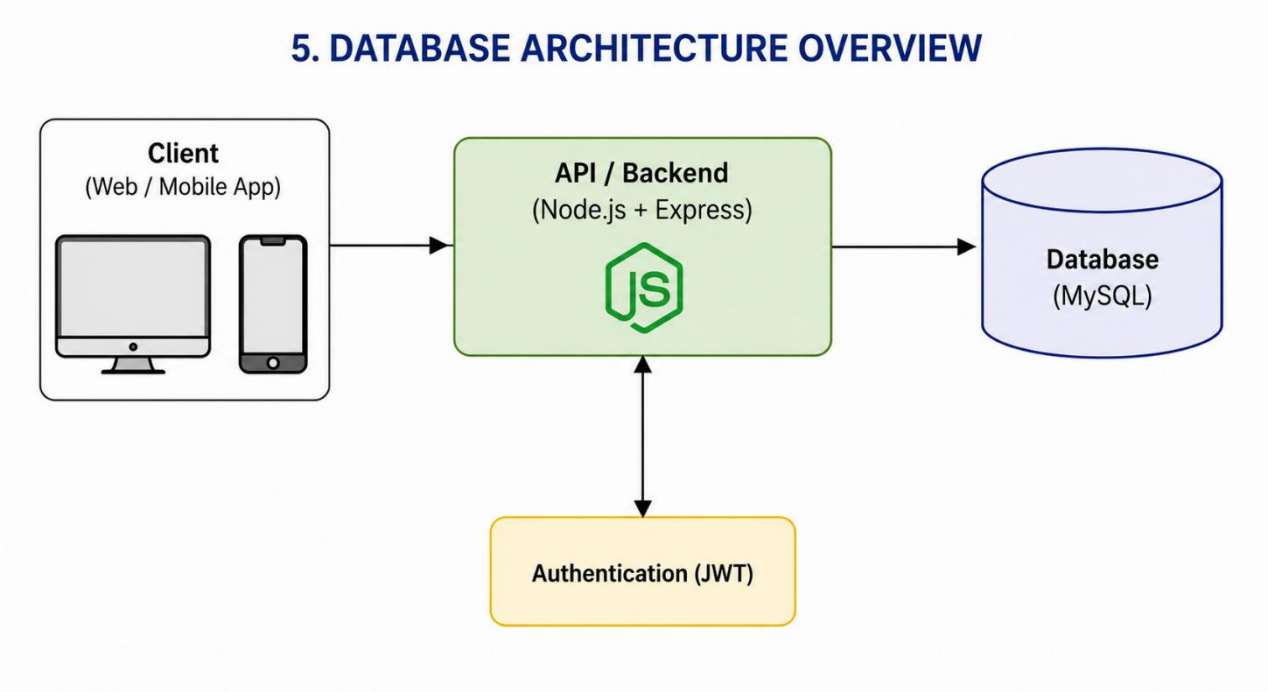
**Description**

The Relationship Diagram generated from the Database Management System (DBMS) visually represents the actual relationships among tables within the MySQL database.The diagram shows how primary keys and foreign keys connect various entities. For example:

- users is linked with patients and doctors.
- patients and doctors are connected through appointments.
- appointments are linked with prescriptions and payments.
- patients can submit reviews for doctors.
- users receive notifications generated by the system.

This DBMS-generated relationship diagram validates the correctness of the database design and confirms that all required relationships have been properly implemented in the database.

**Database Architecture Overview**




**Description**

The Database Architecture Overview illustrates the overall system architecture of MediLink. The architecture follows a three-tier structure:

1. Client Layer (Web and Mobile Applications)
2. API/Backend Layer (Node.js and Express.js)
3. Database Layer (MySQL)

Users interact with the system through web or mobile applications. Client requests are sent to the backend server through RESTful APIs. The backend processes business logic, authentication, appointment management, prescription generation, payment processing, and notification services. Authentication is handled using JSON Web Token (JWT), ensuring secure access control and session management. The backend communicates with the MySQL database to store, retrieve, and update healthcare-related information.This architecture provides scalability, security, maintainability, and efficient data management for the MediLink Healthcare Management System.